

Web Design

Lecture 12 - Web Design Frameworks

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8.1. jQuery

8.2. Bootstrap



jQuery Intro

jQuery Syntax

jQuery Selectors

jQuery Events



- ❧ jQuery is a JavaScript Library.
- ❧ jQuery is a lightweight, "write less, do more", JavaScript library.
- ❧ jQuery takes a lot of common tasks that require many lines of JavaScript code to accomplish, and wraps them into methods that you can call with a single line of code.
- ❧ jQuery also simplifies a lot of the complicated things from JavaScript, like AJAX calls and DOM manipulation.

https://www.w3schools.com/jquery/tryit.asp?filename=tryjquery_hide

What is your impression?



Download the jQuery library from jQuery.com

<http://jquery.com/download/>

Or include jQuery from a CDN, like Google

```
<head>  
<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.7.1/jquery.min.js"></script>  
</head>
```



The jQuery syntax is tailor-made for selecting HTML elements and performing some action on the element(s).

Basic syntax is: **\$(selector).action()**

- ⌘ A \$ sign to define/access jQuery
- ⌘ A (selector) to "query (or find)" HTML elements
- ⌘ A jQuery action() to be performed on the element(s)



Examples:

`$(this).hide()` - hides the current element.

`$("p").hide()` - hides all `<p>` elements.

`$(".test").hide()` - hides all elements with `class="test"`.

`$("#test").hide()` - hides the element with `id="test"`.



This is to prevent any jQuery code from running before the document is finished loading (is ready).

Here are some examples of actions that can fail if methods are run before the document is fully loaded:

- ❌ Trying to hide an element that is not created yet
- ❌ Trying to get the size of an image that is not loaded yet

```
$(document).ready(function(){  
  
    // jQuery methods go here...  
  
});
```




jQuery selectors allow you to select and manipulate HTML element(s).

jQuery selectors are used to "find" (or select) HTML elements based on their name, id, classes, types, attributes, values of attributes and much more. It's based on the existing CSS Selectors, and in addition, it has some own custom selectors.



The jQuery element selector selects elements based on the element name.

You can select all <p> elements on a page like this:

```
$("p")
```

https://www.w3schools.com/jquery/tryit.asp?filename=tryjquery_hide_p



The jQuery #id selector uses the id attribute of an HTML tag to find the specific element.

An id should be unique within a page, so you should use the #id selector when you want to find a single, unique element.

To find an element with a specific id, write a hash character, followed by the id of the HTML element:

```
$("#test")
```

https://www.w3schools.com/jquery/tryit.asp?filename=tryjquery_hide_id



The jQuery .class selector finds elements with a specific class.

To find elements with a specific class, write a period character, followed by the name of the class:

```
$(".test")
```

https://www.w3schools.com/jquery/tryit.asp?filename=tryjquery_hide_class



An event represents the precise moment when something happens.

Examples: moving a mouse over an element, selecting a radio button, clicking on an element

Mouse Events	Keyboard Events	Form Events	Document/Window Events
click	keypress	submit	load
dblclick	keydown	change	resize
mouseenter	keyup	focus	scroll
mouseleave		blur	unload



To assign a click event to all paragraphs on a page, you can do this:

```
$("p").click();
```

The next step is to define what should happen when the event fires. You must pass a function to the event:

```
$("p").click(function(){  
    // action goes here!!  
});
```



Exercise: Thiết kế một trang web gồm các phần sau:

Tiêu đề: Đặt tiêu đề trang là "Bài tập jQuery Events."

Button: Thêm một nút có văn bản "Click me!".

Ô văn bản: Thêm một ô văn bản (input type="text") với một phần tử id là "textInput."

Ảnh: Thêm một ảnh (sử dụng một ảnh bất kỳ) với một phần tử id là "mainImage."

Div: Thêm một div với một phần tử id là "resultDiv."



Yêu cầu tương tác:

Khi người dùng nhấp vào nút "Click me!", hiển thị một thông báo cảnh báo sử dụng jQuery (sử dụng `.click()` hoặc `.on("click")`) với nội dung "Button Clicked!".

Khi người dùng gõ vào ô văn bản và sau đó nhấn Enter (phím ấn xuống), hiển thị giá trị đã nhập vào ô văn bản trong div với id là "resultDiv" (sử dụng `.keypress()` và `.val()`).

Khi người dùng di chuyển con trỏ chuột qua ảnh, thay đổi thuộc tính `src` của ảnh để thay đổi hình ảnh (sử dụng `.mouseenter()` và `.attr()`).

Gợi ý: Để thực hiện bài tập này, bạn cần tạo một tệp HTML mới và bổ sung mã JavaScript sử dụng thư viện jQuery. Sau đó, sử dụng các sự kiện như `.click()`, `.keypress()`, và `.mouseenter()` để thực hiện các chức năng tương ứng.



Hide and Show

Fade

Slide

Animate

Chaining



With jQuery, you can hide and show HTML elements with the `hide()` and `show()` methods:

https://www.w3schools.com/jquery/tryit.asp?filename=tryjquery_hide_slow

```
$("#hide").click(function(){  
    $("p").hide();  
});
```

```
$("#show").click(function(){  
    $("p").show();  
});
```

Syntax:

🔗 `$(selector).hide(speed,callback);`

🔗 `$(selector).show(speed,callback);`



You can also toggle between hiding and showing an element with the toggle() method.

```
$("#button").click(function(){  
    $("#p").toggle();  
});
```

Syntax:

\$(selector).toggle(speed,callback);

https://www.w3schools.com/jquery/tryit.asp?filename=tryjquery_toggle



With jQuery you can fade an element in and out of visibility.

jQuery has the following fade methods:

🔗 `fadeIn()`
https://www.w3schools.com/jquery/tryit.asp?filename=tryjquery_fa_dein

🔗 `fadeOut()`
https://www.w3schools.com/jquery/tryit.asp?filename=tryjquery_fa_deout

🔗 `fadeToggle()`
https://www.w3schools.com/jquery/tryit.asp?filename=tryjquery_fadetoggle

🔗 `fadeTo()`
https://www.w3schools.com/jquery/tryit.asp?filename=tryjquery_fadeto



With jQuery you can create a sliding effect on elements.

jQuery has the following slide methods:

❧ `slideDown()`

❧ `slideUp()`

❧ `slideToggle()`



The jQuery animate() method is used to create custom animations.

Syntax:

```
$(selector).animate({params},speed,callback);
```

```
        $("button").click(function(){  
            $("div").animate({left: '250px'});  
        });
```

- ❧ The required params parameter defines the CSS properties to be animated.
- ❧ The optional speed parameter specifies the duration of the effect. It can take the following values: "slow", "fast", or milliseconds.
- ❧ The optional callback parameter is a function to be executed after the animation completes.



The jQuery animate() method is used to create custom animations.

Syntax:

```

$(selector).animate({params},speed,callback)
    $("button").click(function(){
    $("div").animate({left: '250px'});
    });
    
```

- ⌘ The required params parameter defines the CSS properties to be animated.
- ⌘ The optional speed parameter specifies the duration of the effect. It can take the following values: "slow", "fast", or milliseconds.
- ⌘ The optional callback parameter is a function to be executed after the animation completes.

https://www.w3schools.com/jquery/tryit.asp?filename=tryjquery_animation1_multicss



Chaining allows us to run multiple jQuery methods (on the same element) within a single statement.

To chain an action, you simply append the action to the previous action.

```
$("#p1").css("color", "red").slideUp(2000).slideDown(2000);
```

```
$("#p1").css("color", "red")  
    .slideUp(2000)  
    .slideDown(2000);
```




1. Determine your requirements
2. Browse for jQuery plugins: <https://plugins.jquery.com/>
<https://www.jqueryscript.net/>
3. Download source code (normally .zip files) or link to the plugin (via CDN)
4. Include required CSS
5. Initialize the Plugin
6. Customize Options (Optional)
7. Run and test your site

```
$(document).ready(function(){  
    $("#element").pluginName({  
        // Plugin options (if any)  
    });  
});
```



Step 1: Understand the Plugin's Purpose

- 📖 Slick Carousel is a popular jQuery plugin for creating responsive, swipe-enabled carousels.





Step 2: Include jQuery and Slick Carousel

jQuery:

```
<script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
```

Slick Carousel:

```
<link rel="stylesheet" type="text/css"  
href="https://cdn.jsdelivr.net/npm/slick-carousel@1.8.1/slick/slick.css">
```

```
<script src="https://cdn.jsdelivr.net/npm/slick-carousel@1.8.1/slick/slick.min.js"></script>
```



Step 3: Create the HTML Structure

```
<div class="slick-carousel">  
  <div></div>  
  <div></div>  
  <div></div>  
</div>
```



Step 4: Initialize the Plugin in JavaScript

```
$(document).ready(function(){  
  $('.slick-carousel').slick({  
    autoplay: true, // Enable automatic sliding  
    autoplaySpeed: 2000, // Slide every 2 seconds  
    dots: true, // Add navigation dots  
    arrows: true, // Enable previous/next arrows  
    infinite: true, // Loop slides infinitely  
    slidesToShow: 1, // Show one slide at a time  
    slidesToScroll: 1 // Scroll one slide at a time  
  });  
});
```



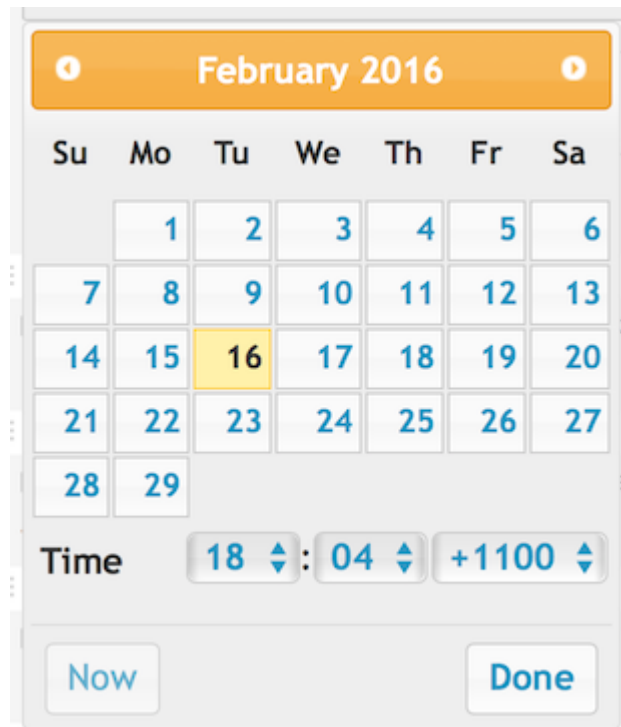
Step 5: Customize Options

The above code includes options like `autoplay`, `autoplaySpeed`, `dots`, and `arrows`. You can adjust these based on your design needs.



Here's another example, this time using the "jQuery UI Datepicker" plugin, which allows users to easily *select dates in a user-friendly calendar* interface. This is often used in forms that require date input.

10 mins for preparation, work in group





Bootstrap

Bootstrap 4 is a newer version of Bootstrap; with new components, faster stylesheet and more responsiveness. However, Internet Explorer 9 and down is not supported.

Bootstrap 5 is the newest version of Bootstrap; with a smooth overhaul. However, Internet Explorer 11 and down is not supported, and **jQuery is replaced with vanilla JavaScript**.

W3.CSS

Tailwind CSS



Try this example:

https://www.w3schools.com/bootstrap4/tryit.asp?filename=trybs_default&stacked=h

What do you think about the source code?



- ❧ Bootstrap is a free front-end framework for faster and easier web development
- ❧ Bootstrap includes HTML and CSS based design templates for *typography, forms, buttons, tables, navigation, modals, image carousels and many other*, as well as optional JavaScript plugins
- ❧ Bootstrap also gives you the ability to easily create responsive designs



Bootstrap 5 (released 2021) is the newest version of Bootstrap; It supports the latest, stable releases of all major browsers and platforms. However, Internet Explorer 11 and down is not supported.

The main differences between Bootstrap 5 and Bootstrap 3 & 4, is that Bootstrap 5 has switched to JavaScript instead of jQuery.



- ❧ Easy to use: Anybody with just basic knowledge of HTML and CSS can start using Bootstrap
- ❧ Responsive features: Bootstrap's responsive CSS adjusts to phones, tablets, and desktops
- ❧ Mobile-first approach: In Bootstrap, mobile-first styles are part of the core framework
- ❧ Browser compatibility: Bootstrap 4 is compatible with all modern browsers (Chrome, Firefox, Internet Explorer 10+, Edge, Safari, and Opera)



🔗 **Include Bootstrap 4 from a CDN**

✂ You just tried this way in the first example

🔗 **Download Bootstrap 4 from getbootstrap.com**

✂ Use an offline-version of Bootstrap

✂ <https://getbootstrap.com/>



My Website [Home](#) [About](#) [Contact](#)

Welcome to My Website

This is a simple hero unit, a simple jumbotron-style component for showcasing content.

[Learn More](#)

Column 1

Lorem ipsum dolor sit amet, consectetur adipiscing elit.

Column 2

Suspendisse vitae nisi urna. Cras euismod lacus id lectus ultricies.

Column 3

Etiam facilisis lorem eget volutpat vehicula.

© 2024 My Website. All rights reserved.

Exercise 1: Add a Responsive Image Gallery



Enhance the webpage by adding a responsive image gallery below the existing content.

- 🔗 Use the Bootstrap grid system to display 6 images.
- 🔗 Arrange the images in two rows with three images per row on medium and larger screens.
- 🔗 On small screens, the images should stack vertically (one image per row).



Add a contact form below the Jumbotron. The form should include the following:

- ✎ A text input for "Name."
- ✎ An email input for "Email."
- ✎ A textarea for "Message."
- ✎ A submit button styled with Bootstrap.

Ensure the form is centered and responsive using a container

Exercise 3: Add a Pricing Table



Below the Jumbotron, add a pricing table with three columns representing different pricing plans (e.g., Basic, Standard, Premium).

- 🔗 Use the Bootstrap grid system to create the columns.
- 🔗 Each column should include: A card with a title (plan name), description, price, and a button.
- 🔗 Style the cards using Bootstrap's card component.



From a list of Bootstrap templates, choose one to analyze

E.g.: <https://startbootstrap.com/themes>

-  Create an empty Google Docs document, name it “Group# - Homepage”
-  Access the home page of the chosen template, capture the whole interface in a single image, paste the image to your document
-  With each single component, capture an image, find the appropriate Bootstrap class, write them in your document

E.g.: <https://startbootstrap.com/previews/agency>



From your reference websites, choose one to analyze.

Write down all the components that can be styled with Bootstrap.